From variational Monte Carlo to superfluid hydrodynamics in microwave-shielded dipolar Bose gases

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We show that, within the manifold where the Jastrow pair factor is frozen to its ground-state form, time-dependent variational Monte Carlo (tVMC) based on a Jastrow–Feenberg wave function maps exactly onto irrotational superfluid hydrodynamics. The theory is then closed by an equation of state extracted from a single variational Monte Carlo calculation of the uniform system; within the local-density approximation, this yields a generalized Gross–Pitaevskii equation with a density-dependent chemical potential $\mu(\rho)$. For a trapped microwave-shielded dipolar Bose gas, the reduced dynamics reproduces the full tVMC breathing oscillations for $N = 8$–128 with below-1% frequency error and about 2% amplitude error, while lowering the computational cost by more than an order of magnitude. Our results establish a first-principles route from correlated many-body wave functions to predictive hydrodynamics for molecular gases, quantum droplets, and supersolids.

Introduction.—Microwave-shielded polar molecules have emerged as a leading platform for dipolar quantum matter. Recent experiments have achieved Bose–Einstein condensation [1, 2], observed shielded collisions [3, 4], and probed few- and many-body phenomena [5–7]. With permanent dipole moments orders of magnitude larger than those of magnetic atoms [8, 9], shielded molecules are expected to exhibit rich collective behavior—from quantum droplets [10, 11] to supersolid-like phases—in a widely tunable setting. Collective oscillations and expansion dynamics serve as primary experimental probes of the underlying interactions in trapped BECs [12–14]; for shielded molecular condensates such measurements remain an open frontier, making first-principles predictions especially timely.

For magnetic-atom dipolar BECs, the extended Gross–Pitaevskii equation (eGPE)—incorporating the non-local dipolar mean field and a perturbative Lee–Huang–Yang (LHY) correction [15]—has successfully described quantum droplets and collective excitations [8, 11]. This framework, however, relies on a Born-approximation coupling constant and treats quantum fluctuations perturbatively; indeed, quantum Monte Carlo calculations have revealed significant deviations from eGPE predictions even for magnetic atoms [16]. For microwave-shielded molecules the situation is more challenging: the strong short-range repulsion ($\sim C_6/r^6$) produces a deep correlation hole that cannot be captured perturbatively, and without it the bare interaction integral diverges. Time-dependent variational Monte Carlo (tVMC) [17–19] can propagate a correlated many-body wave function in real time, but its stochastic sampling at every time step severely limits accessible system sizes and evolution times. What is missing is a controlled coarse-graining that retains the microscopic content of tVMC while recovering the efficiency of hydrodynamic evolution.

Here we establish a first-principles reduction from correlated tVMC to macroscopic hydrodynamics. The key physical ingredient is a separation of scales: short-range pair correlations relax much faster than the global density profile. Under this condition, freezing the two-body Jastrow factor u_2 makes the time-dependent variational principle map exactly onto irrotational superfluid hydrodynamics. We verify this scenario from full tVMC, then close the theory with an equation of state extracted from a single uniform-system VMC calculation within the local-density approximation, yielding a generalized GPE. The reduced theory reproduces the full tVMC breathing dynamics for $N = 8$ and $N = 128$, while extending directly to experimentally relevant large-N regimes beyond direct tVMC.

Model and variational dynamics.—We consider N bosons in a cylindrically symmetric harmonic trap with Hamiltonian

$$\hat{H}(t) = \sum_{i=1}^{N}\left[-\tfrac{1}{2}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \qquad (1)$$

where $V_{\rm tr} = \frac{1}{2}[\omega_\rho(t)^2\rho^2 + \omega_z^2 z^2]$ (natural units $\hbar = m = 1$) and the two-body interaction models microwave-shielded dipolar molecules:

$$v(\mathbf{r}) = \frac{C_3}{r^3}(3\cos^2\theta - 1) + \frac{C_6}{r^6}(\sin^4\theta + w_{12}\sin^2\theta\cos^2\theta). \quad (2)$$

This effective potential, derived from Floquet scattering theory for dual-microwave-dressed molecules [20], is parametrized by coefficients C_3, C_6, and w_{12} fixed by the microwave field configuration. For the representative case studied here we choose $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units, initialize the system in the isotropic trap $\omega_\rho = \omega_z = 1$, and excite radial breathing dynamics with a quench $\omega_\rho : 1 \to 2$ at $t = 0$.

The many-body wavefunction is a truncated Jastrow–Feenberg [21, 22] ansatz:

$$\Psi(\mathbf{X},t) = \exp\Big[\sum_i u_1(\mathbf{r}_i,t) + \sum_{i<j} u_2(r_{ij},\hat{\mathbf{r}}_{ij})\Big], \qquad (3)$$

where the complex one-body field $u_1(\mathbf{r},t)$ controls the evolving density and phase, while u_2 encodes pair correlations. In cylindrical geometry we use a separable parametrization for u_1, and expand u_2 in partial waves as

$$u_2(r,\cos\theta) = u_{\rm cusp}(r) + \sum_{\ell=0,2,4} f_\ell(r)\, P_\ell(\cos\theta), \qquad (4)$$

where $u_{\rm cusp}(r)$ enforces the short-range scattering boundary condition and each f_ℓ is represented by Gaussian basis functions [23]. For the present parameters the $\ell = 2$ anisotropy is weak and the $\ell = 4$ channel is negligible, so the pair correlation is close to isotropic.

Figure 1(a) characterizes the optimized pair correlations. The lowest-order hypernetted-chain estimate $g_2^{\rm HNC}(r,\theta) = e^{2u_2(r,\cos\theta)}$ [24] shows a short-range correlation hole produced by the shielding repulsion, with only weak angular dependence.

The TDVP [25, 26] projects the Schrödinger equation onto the variational tangent space. Writing $O_\alpha = \partial_{\theta_\alpha} \ln\Psi$ and $E_L = (\hat{H}\Psi)/\Psi$, the equations of motion are [17]

$$\sum_\beta S_{\alpha\beta}\,\dot{\theta}_\beta = -i\,F_\alpha, \qquad (5)$$

where the overlap matrix and force vector are

$$S_{\alpha\beta} = \langle \Delta O_\alpha^* \,\Delta O_\beta\rangle_t, \qquad (6)$$
$$F_\alpha = \langle \Delta O_\alpha^* \,\Delta E_L\rangle_t, \qquad (7)$$

with $\Delta A = A - \langle A\rangle_t$ denoting connected fluctuations. Full Monte Carlo and convergence details are given in Ref. [23].

Frozen-u_2 reduction.—The reduction is motivated by a clear separation of scales: the pair correlation u_2 is governed by the interaction potential $v(r)$ and adjusts on the fast time scale $\tau_{\rm pair} \sim r_0^2$ set by the range r_0 of the repulsive shield, whereas the trap quench drives density redistribution on the much slower trap time scale $\sim 1/\omega$. This hierarchy—analogous to the Born–Oppenheimer separation between fast electronic and slow nuclear degrees of freedom [27]—suggests that the pair-correlation sector remains adiabatically pinned to its ground-state form while only the one-body field u_1 evolves. The comparison between full tVMC and this frozen-u_2 approximation confirms the picture quantitatively: Fig. 1(b) shows that the frozen-u_2 evolution closely tracks the full tVMC dynamics for $N = 128$, validating the restriction to fixed u_2.

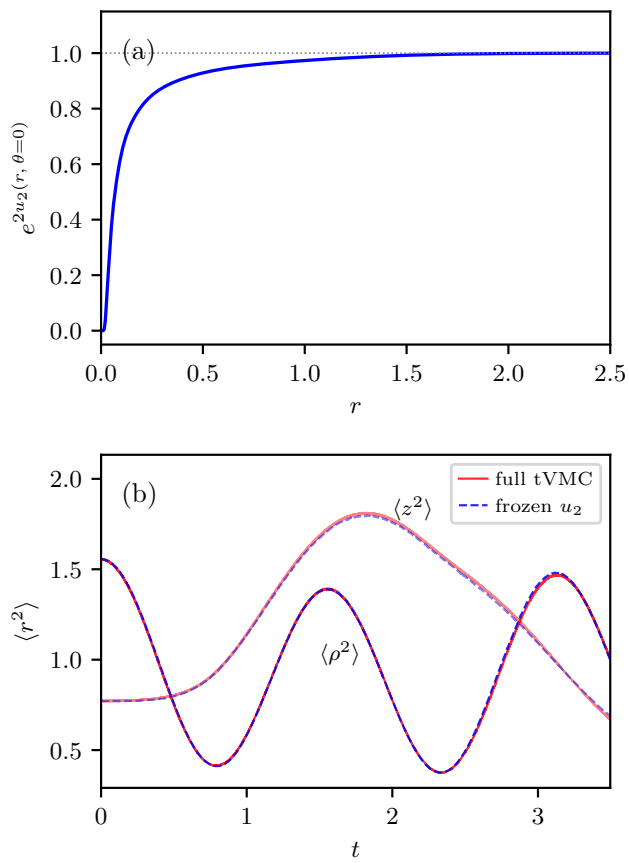

FIG. 1. (a) Lowest-order HNC pair correlation $e^{2u_2(r,\theta=0)}$ for the N=8 ground state obtained by imaginary-time tVMC in the initial isotropic trap $\omega_\rho = \omega_z = 1$, showing a correlation hole at short range due to the shielding repulsion. (b) Breathing dynamics after the quench $\omega_\rho : 1 \to 2$ for N=128: $\langle\rho^2\rangle(t)$ and $\langle z^2\rangle(t)$ from full tVMC (red) and frozen-u_2 tVMC (blue dashed), validating the frozen-u_2 approximation.

When the variational degree of freedom is the continuous field $u_1(\mathbf{r},t)$, the TDVP logarithmic derivative becomes the microscopic density operator:

$$O(\mathbf{r};\mathbf{X}) = \frac{\delta \ln\Psi}{\delta u_1(\mathbf{r})} = \sum_i \delta^{(3)}(\mathbf{r}-\mathbf{r}_i) = \hat{\rho}(\mathbf{r}). \qquad (8)$$

Because u_2 is real and frozen, the many-body phase is purely one-body and therefore irrotational, $\mathbf{v} = \nabla(\mathrm{Im}\, u_1)$; no backflow terms arise. The resulting potential flow precludes vortex formation, so the approximation is tailored to collective modes and quench dynamics that remain irrotational.

Decomposing $u_1 = a + i\theta$ (real amplitude and phase), the TDVP action reduces to the standard hydrodynamic form [23]. The reduction is exact within this frozen-u_2

manifold and yields the continuity and Euler equations:

$$\partial_t \rho + \nabla\cdot(\rho \mathbf{v}) = 0, \tag{9}$$

$$\partial_t \mathbf{v} + \nabla\Big(\tfrac{1}{2}v^2 + V_{\rm tr} + \mu_{u_2}[\rho]\Big) = 0, \tag{10}$$

where $\mu_{u_2}[\rho] = \delta E_{\rm int}[\rho; u_2]/\delta\rho$ is the generally nonlocal chemical-potential functional determined by the frozen correlations.

To obtain a closed local evolution equation, we next invoke the local-density approximation (LDA). The internal energy then takes the form $E_{\rm int} = \int \rho\, \varepsilon(\rho)\, d^3r$, where $\varepsilon(\rho)$ is the energy per particle of the *uniform* system at density ρ with the same frozen u_2. This approximation is valid when the density varies slowly on the scale of the pair-correlation range, $|\nabla\rho|\, r_0/\rho \ll 1$, a condition well satisfied in the bulk of the trapped cloud. The chemical potential becomes a local function,

$$\mu(\rho) = \frac{d[\rho\varepsilon(\rho)]}{d\rho} = \varepsilon + \rho\varepsilon'. \tag{11}$$

The hydrodynamic theory therefore closes in GP form, except that the usual contact term $g\rho$ is replaced by the VMC-derived function $\mu(\rho)$.

The near-stationarity of the pair correlations during the quench is visualized directly in Fig. 2, which shows the change in the lowest-order HNC pair correlation Δe^{2u_2} at four time snapshots. The relative variation remains below $\sim 3\%$ throughout the breathing cycle, even as the one-body density undergoes order-unity redistribution, confirming that u_2 is effectively frozen on the dynamical time scale.

Equation of state and generalized GPE.—We extract $\varepsilon(\rho)$ from a one-time VMC calculation on the uniform system (periodic boundary conditions) with u_2 frozen at its trapped ground-state values (computational details are given in [23]). Over the relevant density range, we fit the results to a polynomial

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + \cdots, \tag{12}$$

yielding the chemical potential $\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + \cdots$.

The striking finding is that $\mu(\rho)$ is nearly linear over the density range relevant to the trapped dynamics (Fig. 3), so the EOS is dominated by the leading term $\mu \approx 2a_1\rho$. This has the same functional form as the standard GPE nonlinearity $g|\psi|^2$, with a *renormalized coupling* $g_{\rm eff} = 2a_1$. Physically, the frozen pair correlations—including the short-range correlation hole visible in Fig. 1(a)—regularize the bare interaction integral

$$\varepsilon_{\rm int}(\rho) = \frac{\rho}{2}\int g_2(\mathbf{r};\rho)\, v(\mathbf{r})\, d^3r, \tag{13}$$

which would diverge for the bare C_3/r^3 potential with $g_2 = 1$. The non-trivial pair structure encoded in u_2 renders this integral finite and maps the realistic anisotropic interaction onto a single effective coupling $g_{\rm eff}$.

This construction differs from phenomenological GP theories in two respects. First, $g_{\rm eff}$ is fixed by the correlated many-body wave function rather than by a scattering-length approximation; for the present parameters we obtain $a_1 \approx 0.134$ from $N = 32$ particles in the periodic simulation cell, converged to within 3% of the $N \to \infty$ extrapolation $a_1^\infty \approx 0.137$ [23]. Second, the small quadratic correction $a_2\rho^2$ captures the residual density dependence of the pair correlations—an effective three-body contribution that emerges naturally from the VMC-derived EOS rather than being inserted ad hoc.

Unlike the eGPE [11, 15], which treats the dipolar interaction through a non-local mean-field kernel supplemented by a perturbative LHY correction, the present approach folds the full interaction—including all non-perturbative pair correlations—into a local function $\mu(\rho)$, extracted once from the uniform-system VMC. This is both computationally simpler (no long-range convolutions at each time step) and microscopically more faithful. The near-linearity of $\mu(\rho)$ is characteristic of the weak-interaction regime ($C_3 = 10^{-3}$); for stronger dipolar coupling, the higher-order polynomial terms become essential, and the framework accommodates this naturally through the full $\mu(\rho)$ nonlinearity in Eq. (14).

The resulting generalized GPE,

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + \mu(|\psi|^2)\Big]\psi, \tag{14}$$

thus takes the standard GP form with the contact coupling replaced by the VMC-derived $\mu(\rho)$; the near-linearity of the EOS explains why a mean-field-like equation captures the correlated dynamics so accurately.

GPE solver.—We solve Eq. (14) where $\psi = \sqrt{\rho}\, e^{i\theta}$ and the Laplacian provides both quantum pressure and flow kinetic energy. The equation is discretized on a cylindrical (r, z) grid exploiting azimuthal symmetry, evaluated with a fourth-order finite-difference Laplacian and time-integrated using a Runge–Kutta scheme under a dispersive CFL condition [23].

The initial state is obtained by imaginary-time propagation to self-consistency: $\psi \to \psi - d\tau\, \hat{H}\psi$, normalized to N particles at each step, until the chemical potential converges to $\delta\mu < 10^{-8}$.

Validation against tVMC.—We benchmark the generalized GPE against frozen-u_2 tVMC at both weak and moderate interaction strength. For $N = 8$, the GPE ground state reproduces the tVMC moments to 0.7% and the energy to 0.15% (Table S1 [23]). The breathing frequency is $\omega_{\rm br} = 4.19 \approx 2\omega_\rho$ in both methods, and the oscillation amplitudes of $\langle\rho^2\rangle$ and $\langle z^2\rangle$ agree to within 1%.

For $N = 128$, where the interaction energy reaches $\sim 21\%$ of the total, the same EOS-based GPE remains accurate. The ground-state moments agree with tVMC

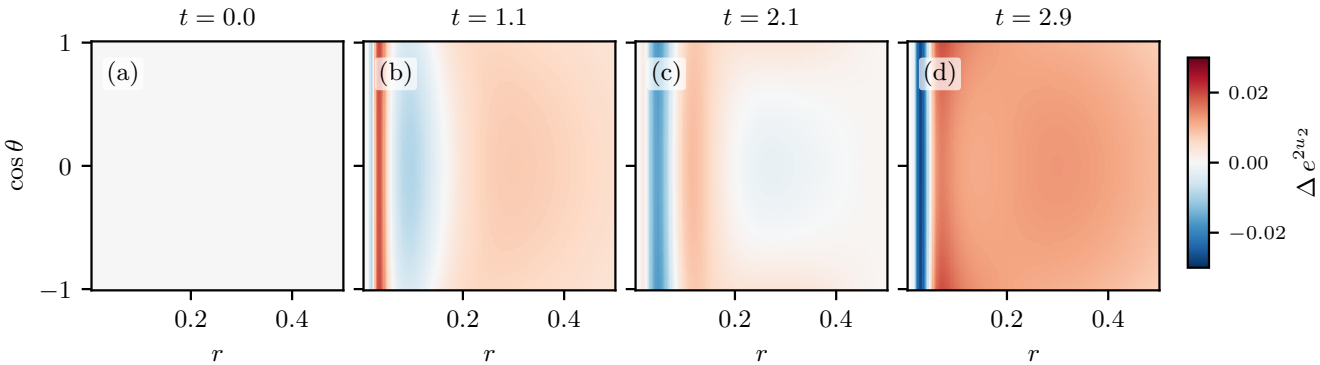

FIG. 2. Change in the lowest-order HNC pair correlation, $\Delta e^{2u_2} = e^{2u_2(t)} - e^{2u_2(0)}$, during breathing dynamics for N=128. The pair correlations remain nearly frozen ($|\Delta e^{2u_2}| \lesssim 0.03$) while the one-body density undergoes order-unity changes.

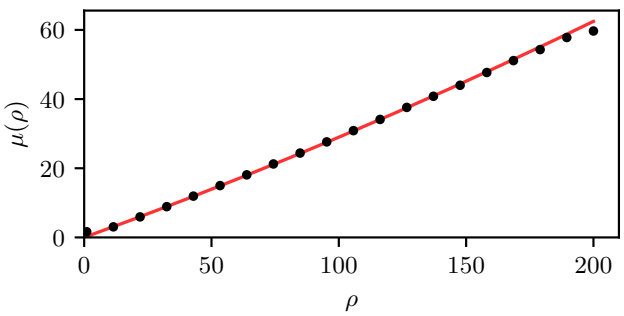

FIG. 3. Chemical potential $\mu(\rho) = d(\rho\varepsilon)/d\rho$ from uniform-system VMC (circles) for C_3=10^{-3}. The line shows the polynomial fit $\mu = 2a_1\rho + 3a_2\rho^2$ used in the GPE. The near-linearity reflects that the frozen pair correlations renormalize the dipolar interaction into an effective contact coupling $g_{\rm eff} = 2a_1$.

to 0.35% for $\langle\rho^2\rangle$ and 1.05% for $\langle z^2\rangle$, while the total energy differs by 2.8%, consistent with a residual LDA error at higher density. The dynamical benchmark in Fig. 4(a) is likewise preserved: both approaches give $\omega_{\rm br} = 4.19$, and the $\langle\rho^2\rangle(t)$ amplitude agrees to about 2%. Notably, although the quench acts only on ω_ρ, the axial moment $\langle z^2\rangle(t)$ also oscillates—driven by nonlinear radial–axial coupling through the interaction energy—and the GPE captures this cross-coupling quantitatively [Fig. 4(a)].

The computational gain is substantial. For N = 128, the GPE evolution, including imaginary-time state preparation, takes $\sim$ 70 minutes on a single CPU core, compared with $\sim$ 6 hours on a modern GPU for the full tVMC.

Large-N prediction.—Having benchmarked the reduction, we use it in a regime inaccessible to direct tVMC. With the same interaction ($C_3 = 10^{-3}$), the generalized GPE predicts sustained breathing oscillations for N = 8192 over six trap periods [Fig. 4(b)], with excellent energy conservation ($|\delta E/E| < 2.5 \times 10^{-4}$).

The N = 8192 system is qualitatively more strongly interacting than the benchmarked cases: the interaction energy comprises $\sim$ 38% of the total (compared with 3.5% for N = 8 and 21% for N = 128), and the peak density reaches $\rho_0 \approx 43$. Despite this tenfold variation in interaction fraction, the breathing frequency remains $\omega_{\rm br} \approx 4.19$ from N = 8 to N = 8192. This robustness is a direct consequence of the near-linearity of $\mu(\rho)$: for a strictly linear chemical potential, the scaling ansatz [28, 29] predicts a breathing frequency that depends only on the trap geometry and is independent of the coupling strength. The quench also drives pronounced axial oscillations: even though ω_z is unchanged, $\langle z^2\rangle$ oscillates with a relative amplitude of $\sim$ 130% [Fig. 4(b)], reflecting strong radial–axial coupling mediated by the interaction at large N.

This particle number falls within the range achieved in recent molecular BEC experiments ($N \sim 10^3$–10^4) [1, 2]. Consequently, the predicted breathing dynamics is directly measurable via experiments, providing a stringent test of the VMC-derived equation of state.

Discussion.—The success of the frozen-u_2 hydrodynamic reduction rests on two pillars: (i) a separation of time scales—pair correlations equilibrate much faster than the macroscopic density redistribution, keeping u_2 nearly stationary—and (ii) the fact that integrating out these correlations yields an EOS dominated by a single renormalized coupling $g_{\rm eff}$.

The near-linearity of $\mu(\rho)$ explains, at a deeper level, why a GPE-like equation captures the correlated dynamics so faithfully even though the microscopic pair correlation $g_2(r) = e^{2u_2}$ differs markedly from the mean-field value $g_2 = 1$. For the hydrodynamic evolution only the functional form of $\mu(\rho)$ matters, not the microscopic origin of its coefficients: once the pair correlations have been integrated into $g_{\rm eff}$, the dynamics is insensitive to the detailed shape of g_2. The value of $g_{\rm eff}$, however, *cannot* be obtained from a mean-field ($g_2 = 1$)

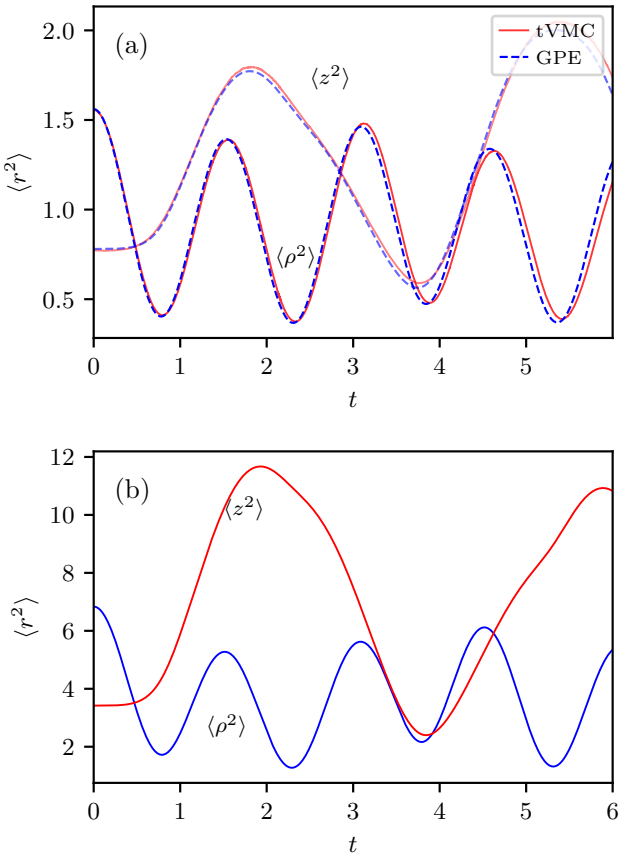

FIG. 4. GPE dynamics from the VMC-derived EOS. (a) N=128 validation: $\langle \rho^2 \rangle(t)$ and $\langle z^2 \rangle(t)$ from frozen-u_2 tVMC (red) and GPE (blue), showing identical breathing frequencies within numerical resolution. (b) N=8192 prediction with the same interaction (C_3=10^{-3}), extrapolating to experimentally relevant system sizes inaccessible to direct tVMC.

treatment, because the bare dipolar integral diverges; the non-trivial correlation hole produced by the shielding repulsion is essential to regularize this integral and fix the coupling. The VMC-derived EOS therefore provides a first-principles route to the effective interaction strength that phenomenological GP theories must take as an external input.

The main quantitative limitation visible here is the $\sim$ 3% energy offset at $N = 128$, which points to gradient corrections beyond the LDA [30]; such terms, of the form $\propto (\nabla\rho)^2/\rho$, could be systematically extracted from VMC at spatially modulated densities and incorporated as a beyond-LDA energy functional. The reduction is expected to break down for quenches faster than the pair-equilibration time ($\tau_{\rm quench} \lesssim \tau_{\rm pair}$), for protocols that nucleate vortices, or near phase transitions (e.g., to a quantum droplet or supersolid) where the pair correlations acquire strong density dependence. Within its domain of validity, the framework extends naturally to other collective modes (scissors, quadrupole), time-dependent trap geometries, and two-component molecular mixtures relevant to current experiments [1, 6, 7].

Conclusion.—We have demonstrated that correlated many-body dynamics from tVMC maps exactly onto superfluid hydrodynamics within the frozen-u_2 manifold, with the required equation of state obtained from a single uniform-system VMC calculation. The resulting generalized GPE reproduces the breathing dynamics of microwave-shielded dipolar bosons from $N = 8$ to $N = 128$ with below-1% frequency error and $\sim 2\%$ amplitude error, and predicts a robust breathing frequency $\omega_{\rm br} \approx 4.19$ that persists from $N = 8$ to $N = 8192$ as the interaction fraction grows from 3.5% to 38%—a prediction directly testable in current molecular BEC experiments. More broadly, the reduction establishes a general strategy for linking correlated variational wave functions to predictive macroscopic dynamics: extract an equation of state from VMC on the uniform system, then evolve via hydrodynamics. This route should be applicable to other quantum fluids with strong short-range correlations, including quantum droplets [10, 16] and dipolar supersolids [8], wherever the pair-correlation sector equilibrates faster than the macroscopic dynamics of interest.

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